

Design a Rate Limiter - Performance Optimization & Monitoring

1. Performance Optimization

Performance optimization is an important aspect of system design.

Area 1: Multi Data Center Setup

- Latency increases for users far from the data center
- Solution:
 - Deploy servers across multiple geographic locations

Example:

- Cloudflare has geographically distributed edge servers

Benefit

None

Route traffic to nearest server → Reduced latency

Area 2: Data Synchronization

- Use **eventual consistency model**

Ensures:

- Data is consistent across distributed systems over time

2. Monitoring

After deploying the rate limiter, analytics must be collected to evaluate effectiveness.

What to Monitor?

✓ Algorithm Effectiveness

- Check if the rate limiting algorithm is working correctly

✓ Rule Effectiveness

- Verify if rules are properly configured

Example Scenarios

Case 1: Rules Too Strict

- Many valid requests are dropped
- Action:

None

Relax rate limiting rules

Case 2: High Traffic (e.g., Flash Sales)

- Rate limiter becomes ineffective
- Action:

None

Use algorithm that supports burst traffic

Example:

- Token Bucket algorithm

3. Key Takeaway

Optimization focuses on:

- Reducing latency using **multi-region deployment**
- Maintaining consistency using **eventual consistency**
- Continuously improving system via **monitoring**